UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Method of Variation of Parameters

Complete study pack for class 011, variation of parameters (the deck's SESSION 8): from the complementary function $y_c = c_1 y_1 + c_2 y_2$ and the Wronskian W, the particular integral $y_p = uy_1 + vy_2$ with $u = -\int \frac{y_2 X}{W} dx$, $v = \int \frac{y_1 X}{W} dx$; every slide and lecture example, all six class-work problems and all four distinct ESA questions (6–7 marks each) solved in full and checked with sympy

Compiled from the PESU Unit 2 deck "Variation of Parameters method" (bucket 011, 18 slides, title slide "SESSION: 8"), its two AV-summary transcripts (T011V, 14.9 min; T011Vb "Continued", 24.5 min), the Unit 1 & 2 class-work sheet (problems 23–28) and eight past ESA papers. Generated 16 October 2026.

· How to use this pack

Topic 1: the method (standard form, three steps, why it works; F1.1–F1.3). Topic 2: the deck's and lectures' examples, including a lecture-only Cauchy equation. Topic 3: trigonometric right-hand sides (CW P23, P25, P26, P28; ESA Jan-24 s1 3a, s2 3a). Topic 4: repeated roots with e^{mx}/x^k (CW P24; ESA Jul-23 3c = Jun-24 3a, May-24 3c) and CW P27. CF cases: class 002 pack; Wronskian: class 001 (E4.3); operator method for $\tan ax$: class 009 (C2.1); Cauchy CF: class 011 Legendre pack. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011V·9	Unit 2 deck 011 “Variation of Parameters method” (SESSION 8), slide 9 (rendered .pptx)
T011V 00:03:26	AV-summary transcript “Variation of Parameters method” (14.9 min, listed first), segment start
T011Vb 00:16:53	AV-summary transcript “Variation of Parameters method_Continued” (24.5 min, second), segment start
CW U2 P25	Unit 1 & 2 class-work sheet, Unit 2 problem 25
ESA Jul-23 3c	Past paper: UE22MA141A July 2023, question 3.c (Jan-24 s1/s2 = UE23MA141A Jan 2024 sets 1/2)

Contents

1. 1 · The method: $y_p = uy_1 + vy_2$ with the Wronskian
2. 2 · Slide and lecture examples: xe^{-x}, $\tan 2x$, $\frac{2}{1+e^x}$, a Cauchy equation
3. 3 · Trigonometric right-hand sides: $\sec ax$, $\csc x$, $e^x \tan x$, $e^{-x}\sec^3 x$
4. 4 · Repeated roots with $X = \frac{e^{mx}}{x^k}$, and the leading-coefficient trap
5. Appendix A · Symbol glossary
6. Appendix B · Sources, coverage and gaps

1 The method: $y_p = uy_1 + vy_2$ with the Wronskian

Variation of parameters finds the particular integral of a second-order linear equation $y'' + py' + qy = X$ whose complementary function $y_c = c_1y_1 + c_2y_2$ is known. The constants c_1, c_2 are replaced by functions $u(x)$, $v(x)$: $y_p = uy_1 + vy_2$ with $u = -\int \frac{y_2 X}{W}dx$, $v = \int \frac{y_1 X}{W}dx$, where $W = y_1y_2' - y_2y_1' \neq 0$ is the Wronskian. The coefficient of y'' must be 1 first. It works for any X (and for variable coefficients once the CF is known); its price is that the CF must be found first and the integrals can be hard.

Key terms and definitions

Method of variation of parameters. A method to find the particular integral of a linear differential equation whose complementary function is known, by replacing the constants c_1, c_2 of $y_c = c_1y_1 + c_2y_2$ by functions $u(x)$, $v(x)$. S011V·4 · T011V 00:00:16 · T011V 00:00:25

Wronskian. $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_2y_1'$; it is non-zero when y_1, y_2 are linearly independent solutions (class 001). S011V·5 · T011V 00:02:35 · T011V 00:03:05

Linearly independent. y_1, y_2 are linearly independent if $c_1y_1 + c_2y_2 \equiv 0$ only for $c_1 = c_2 = 0$; then $c_1y_1 + c_2y_2$ is the CF. (added for clarity) T011V 00:01:48 · T011V 00:02:06

Standard form. $\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = X$: the coefficient of $\frac{d^2y}{dx^2}$ is 1 and X is the right-hand side of this form (divide the whole equation by the leading coefficient first). S011V·4 · S011V·6 · T011V 00:06:06

Parameters u, v. The functions $u = -\int \frac{y_2 X}{W}dx$ and $v = \int \frac{y_1 X}{W}dx$, written without constants of integration. S011V·5 · T011V 00:03:26 · T011V 00:04:18

> *"This method is used to find the particular integral of a linear differential equation whose complimentary function is known. Though this method is general, we will illustrate it by applying to a second order LDE with constant coefficients. Consider a second order linear differential equation $\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = X$ where p and q are constants and X, a function of x. Let its complimentary function be $CF = c_1y_1 + c_2y_2$."* S011V·4

> *"W is the wronskian of y_1 and y_2 and is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. The particular solution of the differential eqn is given by $PI = u.y_1 + v.y_2$ where u and v are given by $u = -\int \frac{y_2.X}{W}dx$ and $v = \int \frac{y_1.X}{W}dx$. Note: The constant of integration is not written in the expressions of u and v."* S011V·5

> *"This method can also be used on differential equation with variable coefficient provided we know to find the general solution of the associated homogeneous equation. The coefficient of $\frac{d^2y}{dx^2}$ in the differential equation must be unity. If it is not unity, make it one by dividing the differential equation by the coefficient."* S011V·6

Formulas

E1.1 · Standard form and CF

ESA ★★★

$$y'' + p\,y' + q\,y = X, \qquad y_c = c_1 y_1 + c_2 y_2$$

GIVES The starting point: y_1, y_2 from the CF.

APPLIES Coefficient of y'' equal to 1.

NOTE Divide by the leading coefficient first (CW P27).

S011V·4 · S011V·6 · T011V 00:00:43 · T011V 00:06:06 · CW U2 P27

E1.2 · Wronskian

ESA ★★★

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1' \neq 0$$

GIVES The denominator of both integrals.

APPLIES y_1, y_2 linearly independent.

NOTE Order matters: $W(y_2, y_1) = -W(y_1, y_2)$, but the PI is unchanged (T011Vb 00:09:20).

S011V·5 · S011V·8 · T011V 00:02:35 · T011V 00:10:22 · ESA Jul-23 3c

E1.3 · The parameters u and v

ESA ★★★

$$u = -\int \frac{y_2\,X}{W}\,dx, \qquad v = \int \frac{y_1\,X}{W}\,dx$$

GIVES The two varying parameters.

APPLIES Every VoP problem.

NOTE Minus sign on u (with y_2); no constants of integration.

S011V·5 · T011V 00:03:26 · T011V 00:04:18 · ESA Jul-23 3c · ESA May-24 3c · ESA Jan-24 s1 3a · ESA Jan-24 s2 3a

E1.4 · The particular integral and the general solution

ESA ★★★

$$y_p = u\,y_1 + v\,y_2, \qquad y = c_1 y_1 + c_2 y_2 + y_p$$

GIVES The PI and, with the CF, the general solution.

APPLIES After E1.3.

NOTE Terms of y_p that are multiples of y_1 or y_2 may be absorbed into c_1, c_2.

S011V·5 · S011V·10 · S011V·17 · T011V 00:14:11 · T011Vb 00:16:16

E1.5 · Why: the two conditions on u', v' ESA ★

$$u'y_1 + v'y_2 = 0, \qquad u'y_1' + v'y_2' = X$$
$$\Rightarrow\ u' = -\frac{y_2 X}{W}, \quad v' = \frac{y_1 X}{W}$$

GIVES E1.3 by Cramer's rule.

APPLIES Derivation; not on the slides.

NOTE The first condition is a free choice that keeps y_p' free of u', v'; the second is the equation itself.

ADDED FOR CLARITY S011V·5

Graphs and diagrams

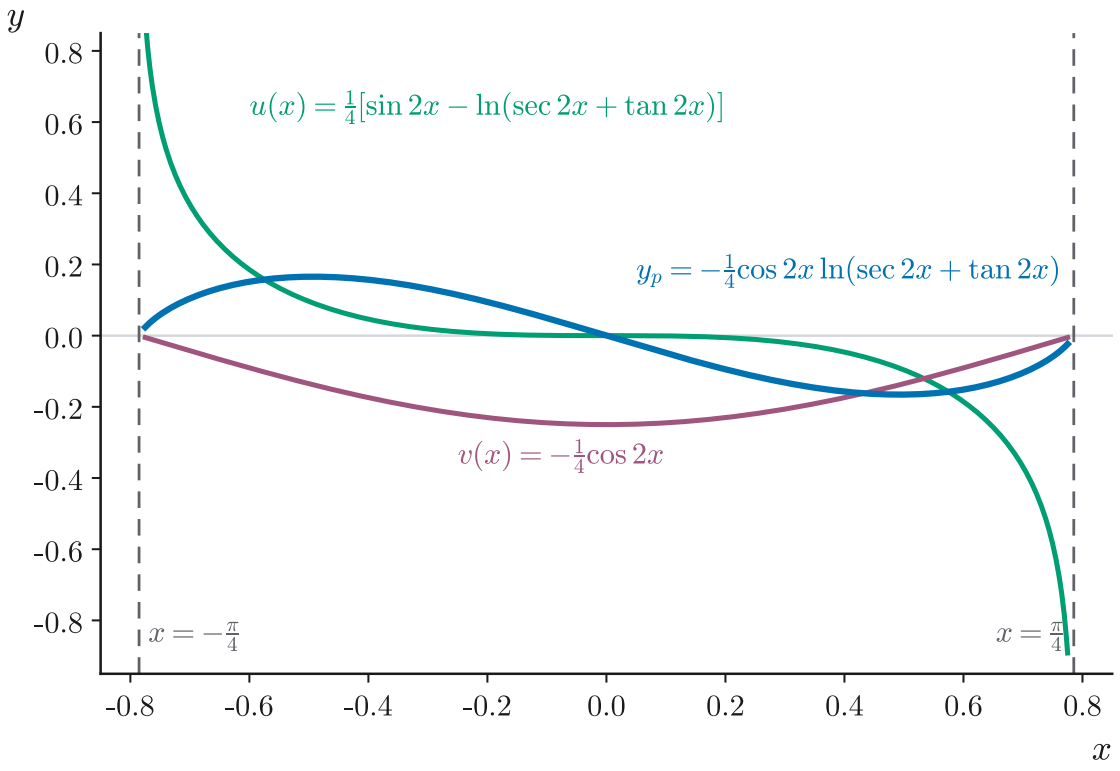
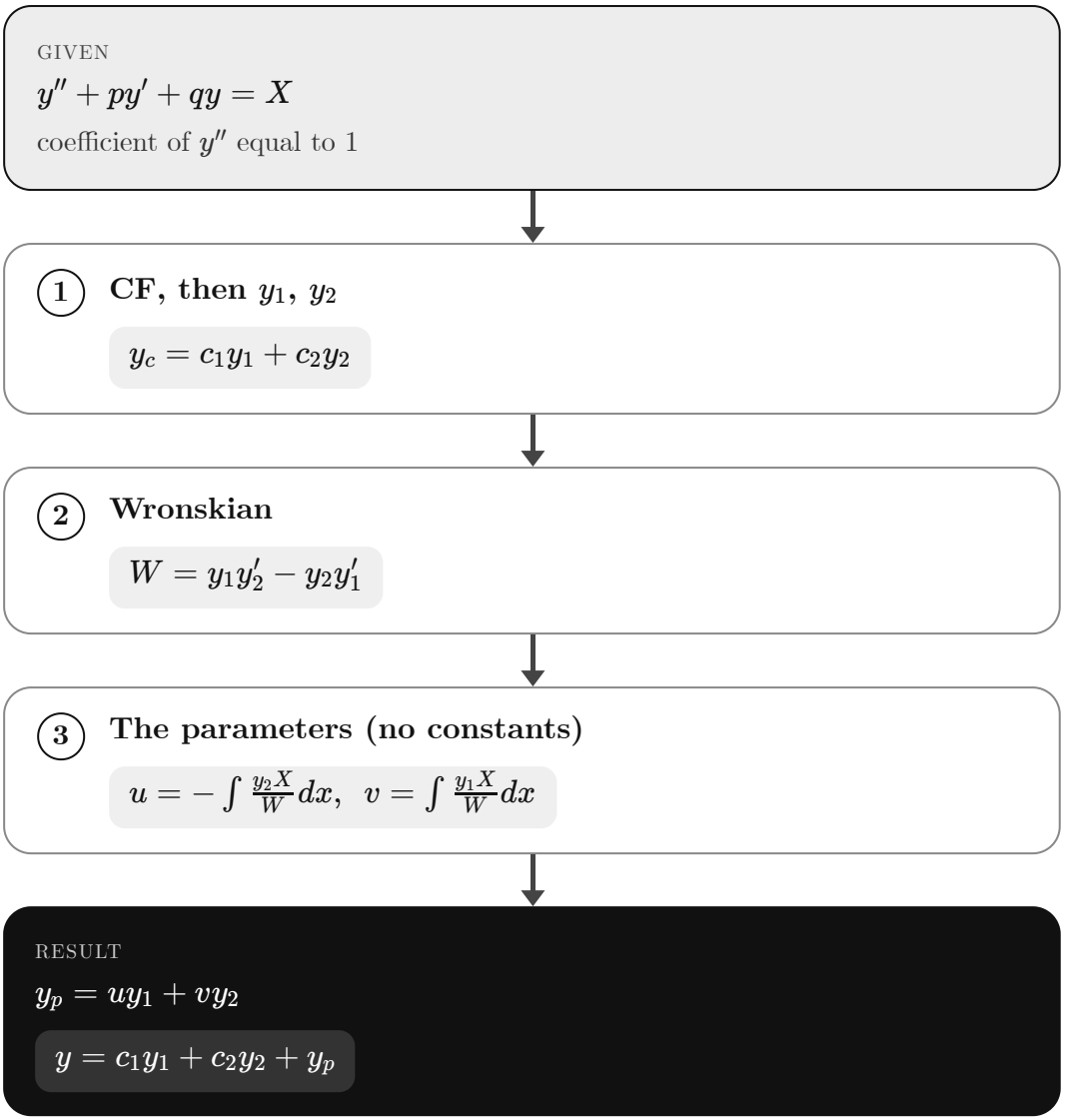


Figure G1.1. The parameters vary: $u(x)$, $v(x)$ and y_p for $(D^2+4)y = \tan 2x$. ESA ★★

ADDED FOR CLARITY S011V·12 · S011V·13

How to read it. On $-\frac{\pi}{4} < x < \frac{\pi}{4}$. Green: $u = \frac{1}{4}[\sin 2x - \log(\sec 2x + \tan 2x)]$; purple: $v = -\frac{\cos 2x}{4}$. In the CF they would be constants (horizontal lines); here they change with x, which is the idea of the method. Blue: $y_p = u\cos 2x + v\sin 2x = -\frac{1}{4}\cos 2x \log(\sec 2x + \tan 2x)$, which stays bounded although u blows up at the ends.

Flowcharts

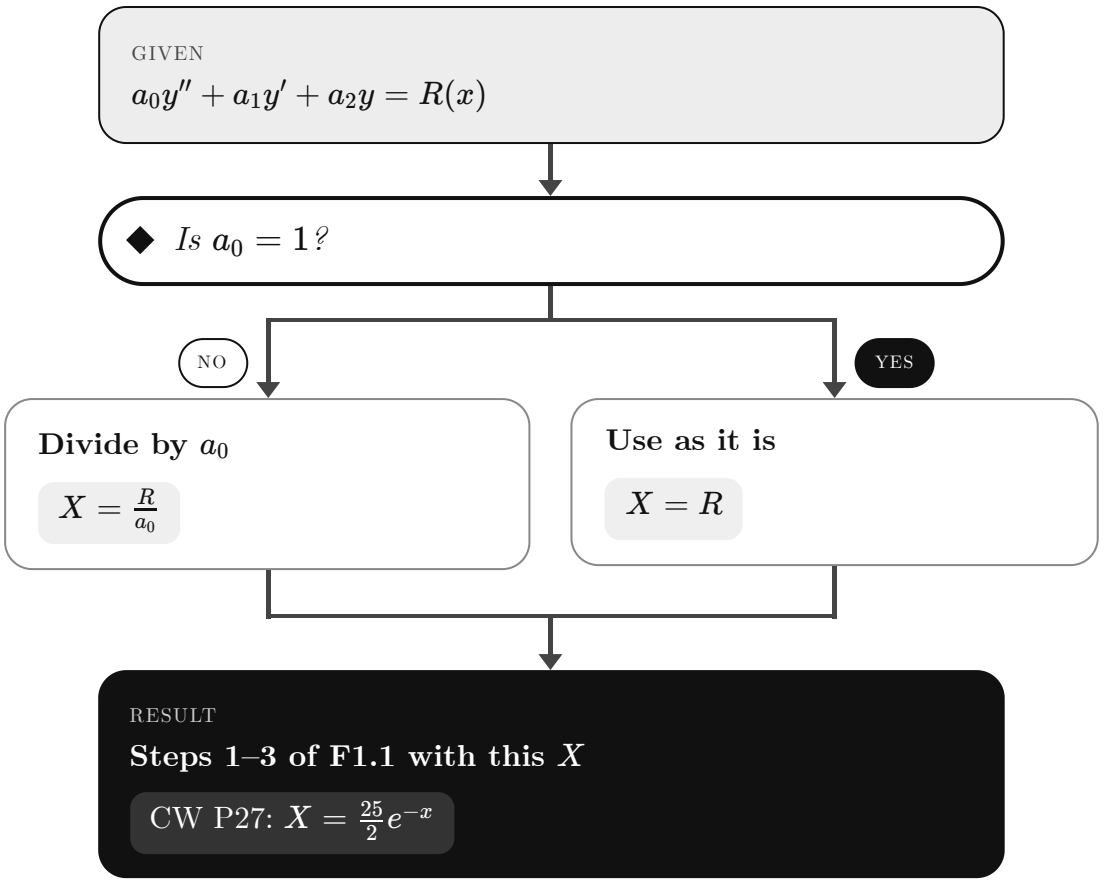


Remark. Terms of y_p that are multiples of y_1 or y_2 may be absorbed into c_1, c_2.

Exam trap. Minus sign on u; u has y_2 inside, v has y_1.

Figure F1.1. Variation of parameters in three steps (slides 011V·4–5) ESA ★★★

S011V·4 · S011V·5 · T011V 00:08:55 · T011Vb 00:08:20



Remark. The CF (AE roots) is the same either way; only X changes.

Figure F1.2. Before you start: the standard form (slide 011V·6) ESA ★★★

S011V·6 · T011V 00:06:06 · CW U2 P27

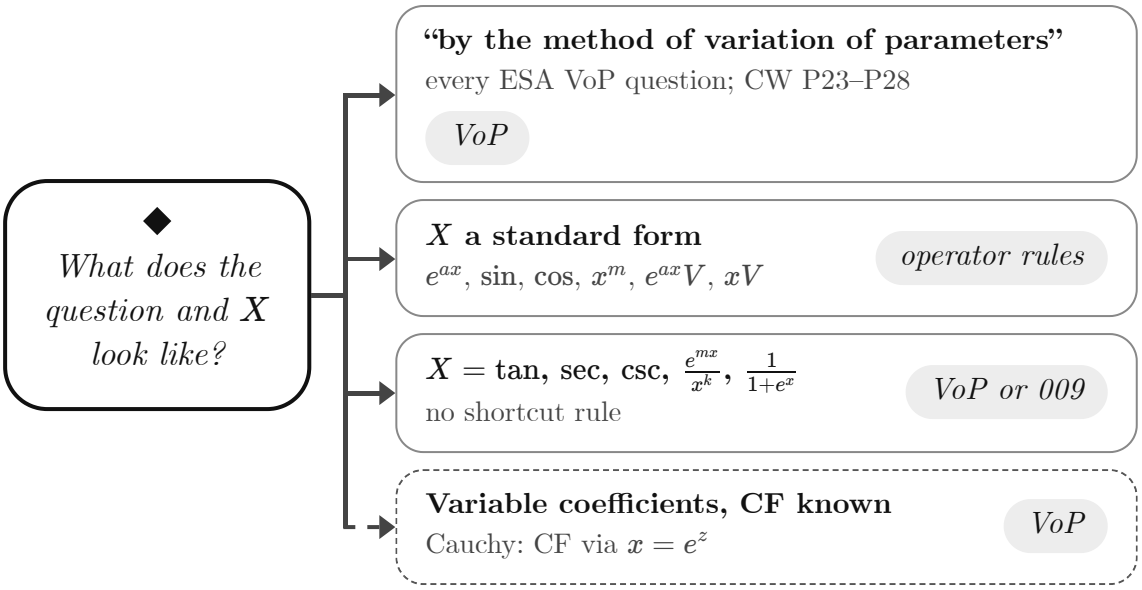


Figure F1.3. Which method for the PI? ESA ★★★

ADDED FOR CLARITY T011V 00:08:02 · T011V 00:08:19 · ESA Jul-23 3c · CW U2 P23

Variation of parameters or the operator rules?

	Operator rules (classes 003–006, 009)	Variation of parameters (this pack)
Needs the CF first?	no (PI alone)	yes: y_1, y_2 and W
Which X?	standard forms; 009 for any X via $\frac{1}{D-m}$	any X
Variable coefficients	only after a substitution	yes, if the CF is known
Main difficulty	choosing the rule; failure cases	the integrals
Use when	question says "solve"	question says "by the method of variation of parameters"

ADDED FOR CLARITY T011V 00:06:31 · T011V 00:08:02 · S011V·6

Important points and why

1. The method gives only the particular integral, and it needs the complementary function first: "the complementary function is absolutely required to apply this method". ESA ★★★ S011V·4 · T011V 00:00:25 · T011V 00:06:42 · T011Vb 00:01:12

2. Three steps: (1) CF, read off y_1, y_2; (2) Wronskian $W = y_1y_2' - y_2y_1'$; (3) $u = -\int \frac{y_2X}{W}dx$, $v = \int \frac{y_1X}{W}dx$, $y_p = uy_1 + vy_2$ (Figure F1.1). ESA ★★★ T011V 00:08:55 · T011Vb 00:01:29 · T011Vb 00:08:20 · S011V·7 · S011V·8 · S011V·9
3. Make the coefficient of $\frac{d^2y}{dx^2}$ equal to 1 before reading off X; otherwise u and v are wrong by that factor. ESA ★★★ S011V·6 · T011V 00:06:06 · CW U2 P27
4. No constants of integration in u, v: "the constant of integration gets absorbed with the constants of the complementary function". ESA ★★★ S011V·5 · T011V 00:04:18 · T011Vb 00:07:06
5. Use it when the question says "by the method of variation of parameters" (every ESA VoP question does), or when X is not a standard form ($\tan$, $\sec$, $\csc$, $\frac{e^{mx}}{x}$, $\frac{1}{1+e^x}$, ...). ESA ★★★ T011V 00:08:19 · ESA Jul-23 3c · ESA Jan-24 s1 3a · CW U2 P23
6. It is general: it also works for variable coefficients when the CF is known, and for higher order; only second order is taught. ESA ★ S011V·6 · T011V 00:05:09 · T011V 00:05:39 · T011Vb 00:16:33
7. Demerits (lecture): it needs the CF, and the integrals can be very difficult. ESA ★★ T011V 00:06:31 · T011V 00:06:55 · T011Vb 00:04:19
8. Swapping y_1 and y_2 changes the sign of W but not the final PI. ESA ★★ T011Vb 00:09:02 · T011Vb 00:09:20

Why are u' and v' given by those formulas?
Try $y_p = uy_1 + vy_2$ and impose $u'y_1 + v'y_2 = 0$; then $y_p' = uy_1' + vy_2'$ and substituting into $y'' + py' + qy = X$ leaves $u'y_1' + v'y_2' = X$ (because y_1, y_2 solve the homogeneous equation). Cramer's rule on the two equations gives $u' = -\frac{y_2X}{W}$, $v' = \frac{y_1X}{W}$.
ADDED FOR CLARITY S011V·5

Why must the coefficient of y'' be 1?
The derivation substitutes into $y'' + py' + qy = X$; the second condition is $u'y_1' + v'y_2' = X$ only for that form. For $2y'' + \cdots = R$ the right-hand side to use is $X = \frac{R}{2}$.
ADDED FOR CLARITY S011V·6 · T011V 00:06:06

Why may constants of integration be dropped?
A constant k in u adds ky_1 to y_p, which is already in the CF c_1y_1; the same for v and y_2.
T011V 00:04:18 · S011V·5

Why is $W \neq 0$ needed?
W is the determinant of the system for u', v'; it is non-zero exactly because y_1, y_2 are linearly independent.
T011V 00:03:05 · S011V·5

Exam trap. Forgetting the minus sign in $u = -\int \frac{y_2X}{W}dx$, or pairing y_1 with u's integral: u goes with y_2 inside, v with y_1. ADDED FOR CLARITY S011V·5

Exam trap. Sign of W: for e^{3x}, e^{-x} it is $-4e^{2x}$, negative. Carry the sign through u and v. S011V·8 · S011V·14

Exam trap. "Linearly independent if $c_1y_1 + c_2y_2 \neq 0$" (lecture) is loose: it means $c_1y_1 + c_2y_2 \equiv 0$ forces $c_1 = c_2 = 0$. ADDED FOR CLARITY T011V 00:02:06

SHORT-NOTE RECAP

Variation of parameters gives the PI of $y'' + py' + qy = X$ once the CF $c_1y_1 + c_2y_2$ is known. Make the coefficient of y'' 1. Wronskian $W = y_1y_2' - y_2y_1' \neq 0$. $u = -\int \frac{y_2X}{W}dx$, $v = \int \frac{y_1X}{W}dx$ (no constants), $y_p = uy_1 + vy_2$, $y = y_c + y_p$. Works for any X and, with a known CF, for variable coefficients. Demerits: needs the CF; integrals can be hard.

2 Slide and lecture examples: xe^{-x}, $\tan 2x$, $\frac{2}{1+e^x}$, a Cauchy equation

The deck works three examples in the same three steps (CF and y_1, y_2 → Wronskian → u, v, PI): $y'' - 2y' - 3y = xe^{-x}$ (by parts), $(D^2+4)y = \tan 2x$ (the sec integral) and $(D^2-1)y = \frac{2}{1+e^x}$ (partial fractions and the substitution $t = 1 + e^{-x}$). Slide 011V·13's boxed answer drops a factor $\cos 2x$. The second lecture also solves the Cauchy equation $x^2y'' - 3xy' + 3y = 12x^4$ by the method, working in $z = \log x$ (transcript only).

Formulas

E2.1 · Integrals of slides 011V·12–16 ESA ★★

$$\int \sec 2x\,dx = \tfrac{1}{2}\log(\sec 2x + \tan 2x), \qquad \int \frac{dx}{e^x(1+e^x)} = -e^{-x} + \log(1+e^{-x})$$

$$\int \frac{e^x}{1+e^x}dx = \log(1+e^x), \qquad \int xe^{-4x}dx = -\frac{e^{-4x}}{4}\left(x + \frac{1}{4}\right)$$

GIVES The four integrals the deck needs.

APPLIES $\tan 2x$, $\frac{2}{1+e^x}$, xe^{-x}.

NOTE $\frac{1}{e^x(1+e^x)} = \frac{1}{e^x} - \frac{1}{1+e^x}$ and $\frac{1}{1+e^x} = \frac{e^{-x}}{1+e^{-x}}$.

S011V·9 · S011V·12 · S011V·15 · S011V·16 · S011V·17 · T011Vb 00:05:21 · T011Vb 00:11:06 · T011Vb 00:12:55

E2.2 · Result: $(D^2+4)y = \tan 2x$ ESA ★★

$$y_p = -\frac{1}{4}\cos 2x\,\log(\sec 2x + \tan 2x)$$

GIVES The PI (slide 011V·13's line above the box; CW P28).

APPLIES $u = \frac{1}{4}[\sin 2x - \log(\sec 2x + \tan 2x)]$, $v = -\frac{\cos 2x}{4}$.

NOTE Slide 011V·13's box omits $\cos 2x$.

S011V·13 · CW U2 P28 · T011Vb 00:07:06

E2.3 · Result: $(D^2-1)y = \frac{2}{1+e^x}$ ESA ★★

$$y_p = e^x\log(1+e^{-x}) - e^{-x}\log(1+e^x) - 1$$

GIVES The PI (slide 011V·17).

APPLIES $u = -e^{-x} + \log(1+e^{-x})$, $v = -\log(1+e^x)$.

NOTE The -1 is not a CF term: keep it.

S011V·17 · T011Vb 00:15:48

Graphs and diagrams

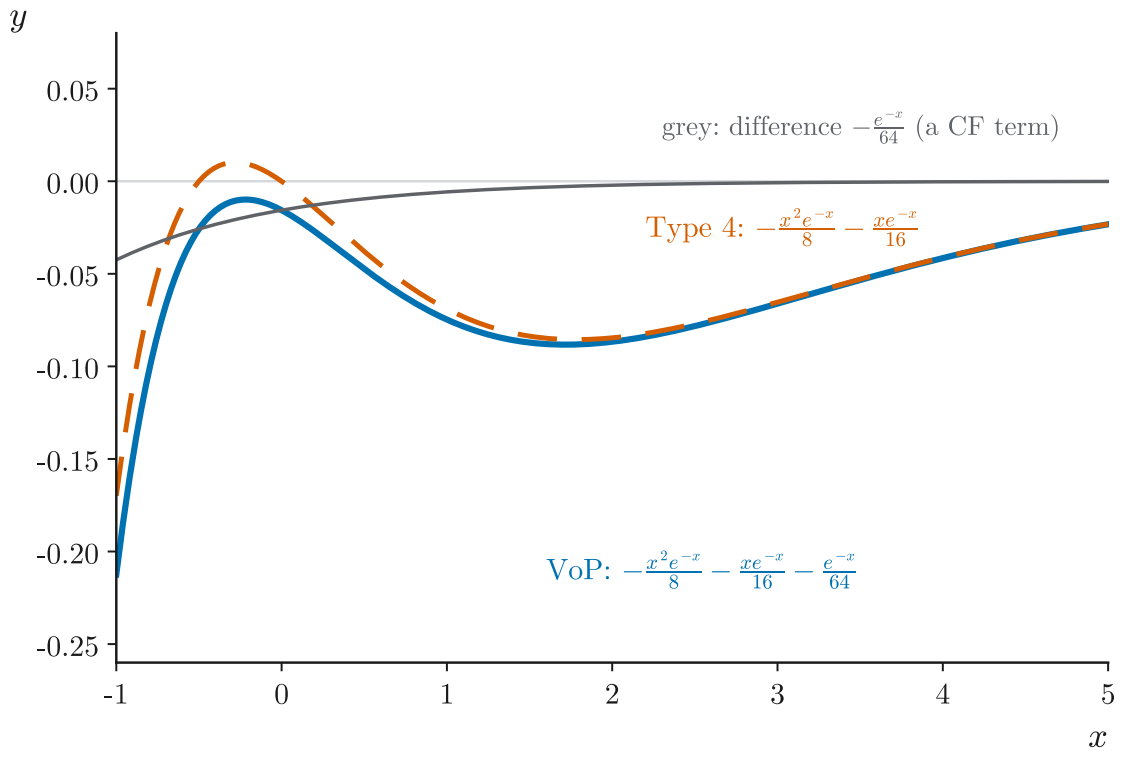


Figure G2.1. Slide 011V·10: two particular integrals differ by a CF term. ESA ★★

ADDED FOR CLARITY S011V·10 · T011V 00:08:02

How to read it. Blue: the VoP PI of slide 011V·10, $-\frac{x^2e^{-x}}{8} - \frac{xe^{-x}}{16} - \frac{e^{-x}}{64}$. Vermilion dashed: the Type 4 PI $-\frac{x^2e^{-x}}{8} - \frac{xe^{-x}}{16}$. Grey: their difference $-\frac{e^{-x}}{64}$, a multiple of $y_2 = e^{-x}$, so both are correct PIs.

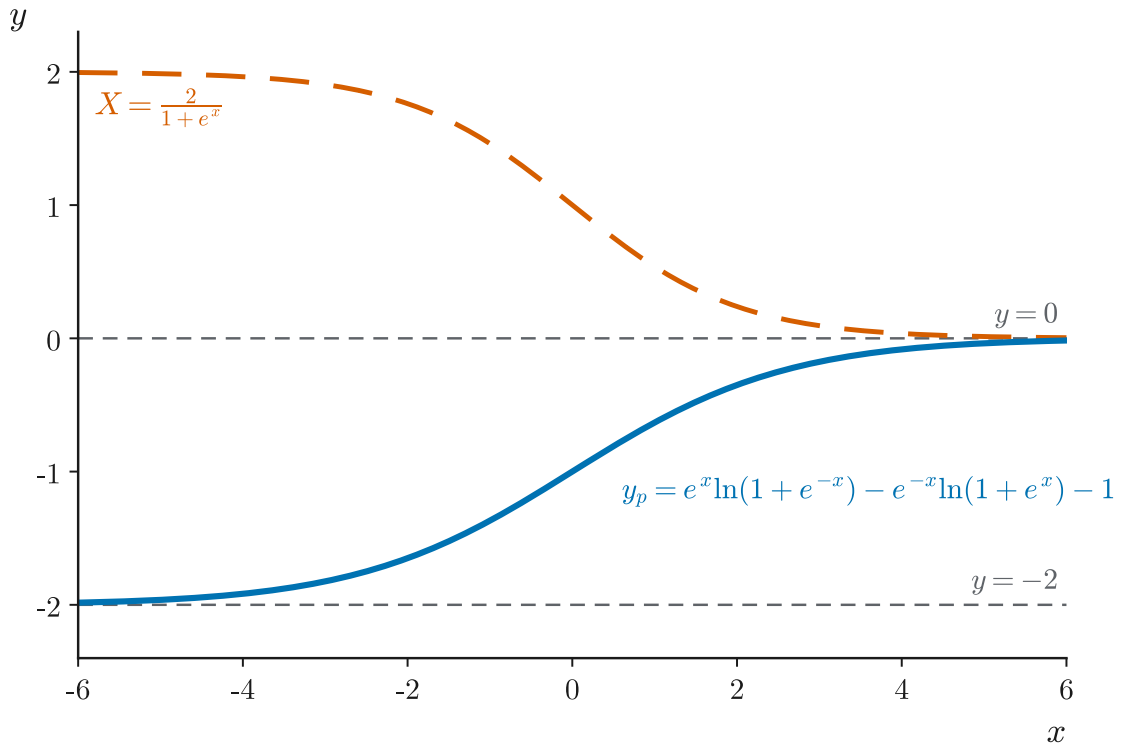
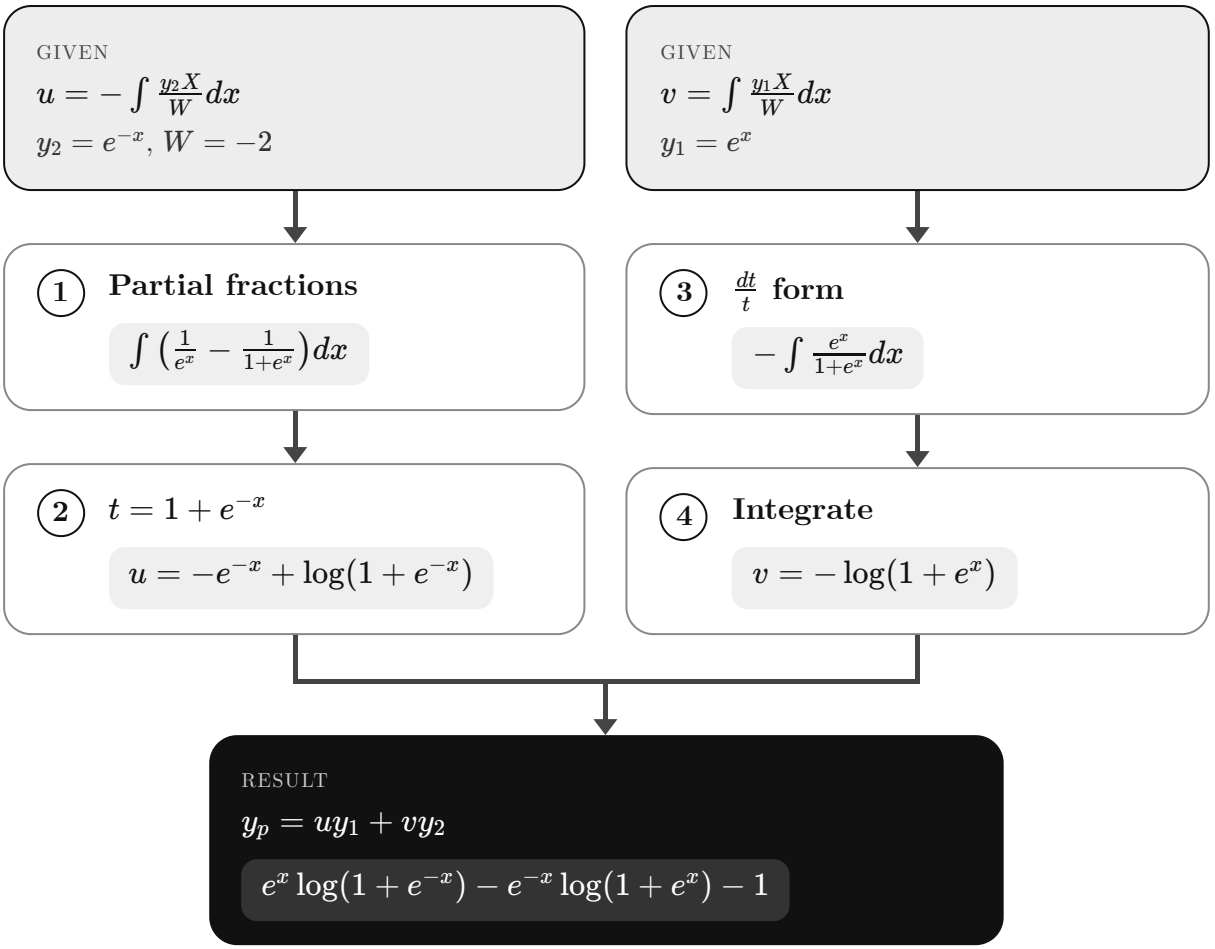


Figure G2.2. $(D^2-1)y = \frac{2}{1+e^x}$: the PI follows $-X$ where X changes slowly. ESA ★★

ADDED FOR CLARITY S011V·17

How to read it. Vermilion dashed: $X = \frac{2}{1+e^x}$, from 2 (left) to 0 (right). Blue: $y_p = e^x \log(1+e^{-x}) - e^{-x}\log(1+e^x) - 1$. For very negative x, $y_p \to -2$ (the PI of $y''-y=2$); for large x, $y_p \to 0$. Grey: the levels -2 and 0.

Flowcharts



Remark. Worked in full as W2.3.

Figure F2.1. Slides 011V·14–17: $(D^2 - 1)y = \frac{2}{1+e^x}$ ESA ★★

S011V·14 · S011V·15 · S011V·16 · S011V·17

Important points and why

1. $y'' - 2y' - 3y = xe^{-x}$ could be done by the Type 4 rule; by VoP $W = -4e^{2x}$, u by parts, $v = -\frac{x^2}{8}$; the two PIs differ by $-\frac{e^{-x}}{64}$, a CF term. ESA ★★ S011V·7 · S011V·10 · T011V 00:08:02
2. For $\tan 2x$ write $\frac{\sin^2 2x}{\cos 2x} = \sec 2x - \cos 2x$ before integrating. ESA ★★ S011V·12 · T011Vb 00:04:58
3. For $\frac{2}{1+e^x}$: partial fractions $\frac{1}{e^x(1+e^x)} = \frac{1}{e^x} - \frac{1}{1+e^x}$, then multiply by e^{-x} to get $\frac{e^{-x}}{1+e^{-x}}$, a $\frac{dt}{t}$ form. ESA ★★ S011V·15 · S011V·16 · T011Vb 00:11:06 · T011Vb 00:12:10
4. For a Cauchy equation, first put $x = e^z$; then do VoP in z and return to x at the end (lecture only). ESA ★ (TRANSCRIPT ONLY) T011Vb 00:17:57 · T011Vb 00:20:11 · T011Vb 00:23:01

Exam trap. Slide 011V·13's box writes $y_p = -\frac{1}{4}\log(\sec 2x + \tan 2x)$: the factor $\cos 2x$ is missing. The correct PI is $-\frac{1}{4}\cos 2x \log(\sec 2x + \tan 2x)$ (the slide's own previous line; CW P28). S011V·13 · CW U2 P28

Exam trap. In $(D^2-1)y=\frac{2}{1+e^x}$ the constant -1 in y_p is not a CF term (the CF is $c_1e^x+c_2e^{-x}$): do not drop it. ADDED FOR CLARITY S011V·17

Worked examples

W2.1 · $y''-2y'-3y=xe^{-x}$ (slides 011V·7–10) ESA ★★

GIVEN Find the particular integral by the method of variation of parameters: $(D^2-2D-3)y=xe^{-x}$.

STEP 1 Step 1: AE $m^2-2m-3=0$, $m=3,-1$. $y_c=c_1e^{3x}+c_2e^{-x}$; $y_1=e^{3x}$, $y_2=e^{-x}$; $X=xe^{-x}$.

STEP 2 Step 2: $W=e^{3x}(-e^{-x})-e^{-x}(3e^{3x})=-4e^{2x}$.

STEP 3 $u=-\int\frac{e^{-x}\cdot xe^{-x}}{-4e^{2x}}dx=\frac{1}{4}\int xe^{-4x}dx=\frac{1}{4}\left[-\frac{xe^{-4x}}{4}-\frac{e^{-4x}}{16}\right]=-\frac{e^{-4x}}{16}\left(x+\frac{1}{4}\right)$ (by parts).

STEP 4 $v=\int\frac{e^{3x}\cdot xe^{-x}}{-4e^{2x}}dx=-\frac{1}{4}\int x\,dx=-\frac{x^2}{8}$.

STEP 5 $y_p=uy_1+vy_2=-\frac{e^{-x}}{16}\left(x+\frac{1}{4}\right)-\frac{x^2e^{-x}}{8}$.

ANSWER $$\boxed{y_p=-\frac{x^2e^{-x}}{8}-\frac{xe^{-x}}{16}-\frac{e^{-x}}{64};\ y=c_1e^{3x}+c_2e^{-x}+y_p}$$

CHECK Slide 011V·10 agrees. sympy: residual 0; dsolve gives $C_2e^{3x}+(C_1-\frac{x^2}{8}-\frac{x}{16})e^{-x}$, i.e. the $-\frac{e^{-x}}{64}$ is absorbed into c_2; the Type 4 rule gives the same. Figure G2.1.

S011V·7 · S011V·8 · S011V·9 · S011V·10 · T011V 00:07:10 · T011V 00:14:03

W2.2 · $(D^2+4)y=\tan 2x$ (slides 011V·11–13; CW P28) ESA ★★

GIVEN Find the PI by variation of parameters (CW P28: solve) of $\frac{d^2y}{dx^2}+4y=\tan 2x$.

STEP 1 AE $m^2+4=0$, $m=\pm 2i$. $y_c=c_1\cos 2x+c_2\sin 2x$; $y_1=\cos 2x$, $y_2=\sin 2x$.

STEP 2 $W=\cos 2x\cdot 2\cos 2x-\sin 2x(-2\sin 2x)=2$.

STEP 3 $u=-\int\frac{\sin 2x\tan 2x}{2}dx=-\frac{1}{2}\int\frac{1-\cos^2 2x}{\cos 2x}dx=-\frac{1}{2}\int(\sec 2x-\cos 2x)dx=$ $-\frac{1}{2}\left[\frac{\log(\sec 2x+\tan 2x)}{2}-\frac{\sin 2x}{2}\right]=\frac{1}{4}[\sin 2x-\log(\sec 2x+\tan 2x)]$.

STEP 4 $v=\int\frac{\cos 2x\tan 2x}{2}dx=\frac{1}{2}\int\sin 2x\,dx=-\frac{\cos 2x}{4}$.

STEP 5 $y_p=\frac{1}{4}[\sin 2x-\log(\sec 2x+\tan 2x)]\cos 2x-\frac{\sin 2x\cos 2x}{4}$: the $\sin 2x\cos 2x$ terms cancel.

ANSWER $$\boxed{y=c_1\cos 2x+c_2\sin 2x-\frac{1}{4}\cos 2x\,\log(\sec 2x+\tan 2x)}$$

CHECK **Slide 011V·13's box drops $\cos 2x$**; substituting it back fails, the answer above leaves residual 0 (sympy). Matches CW P28 and the operator method with $a=2$ (class 009, W2.1). T011Vb 00:06:49 says $v=-\frac{\cos 2x}{2}$; it is $-\frac{\cos 2x}{4}$. Figure G1.1.

S011V·11 · S011V·12 · S011V·13 · T011Vb 00:00:53 · T011Vb 00:07:06 · CW U2 P28

W2.3 · $(D^2-1)y = \frac{2}{1+e^x}$ (slides 011V·14–17) ESA ★★

GIVEN Find the general solution by the method of variation of parameters: $(D^2-1)y = \frac{2}{1+e^x}$.

STEP 1 AE $m^2-1=0$, $m=\pm 1$. $y_c = c_1e^x + c_2e^{-x}$; $y_1 = e^x$, $y_2 = e^{-x}$. $W = e^x(-e^{-x}) - e^{-x}e^x = -2$.

STEP 2 $u = -\int \frac{e^{-x}\cdot\frac{2}{1+e^x}}{-2}dx = \int \frac{dx}{e^x(1+e^x)} = \int\left(\frac{1}{e^x} - \frac{1}{1+e^x}\right)dx$.

STEP 3 $\int e^{-x}dx = -e^{-x}$; $\int \frac{dx}{1+e^x} = \int \frac{e^{-x}}{1+e^{-x}}dx$, $t = 1+e^{-x}$, $dt = -e^{-x}dx$: $= -\log(1+e^{-x})$.

STEP 4 So $u = -e^{-x} + \log(1+e^{-x})$.

STEP 5 $v = \int \frac{e^x\cdot\frac{2}{1+e^x}}{-2}dx = -\int \frac{e^x}{1+e^x}dx = -\log(1+e^x)$.

STEP 6 $y_p = e^x[-e^{-x} + \log(1+e^{-x})] + e^{-x}[-\log(1+e^x)]$.

ANSWER $\boxed{y = c_1e^x + c_2e^{-x} + e^x\log(1+e^{-x}) - e^{-x}\log(1+e^x) - 1}$

CHECK Slide 011V·17 agrees (it stops at "$y = y_c + y_p$"). sympy: u', v' match the integrands, residual 0. As $x\to-\infty$, $X\to 2$ and $y_p\to -2$; as $x\to\infty$, $y_p\to 0$. Figure G2.2, Figure F2.1.

S011V·14 · S011V·15 · S011V·16 · S011V·17 · T011Vb 00:07:39 · T011Vb 00:15:48

W2.4 · $x^2y'' - 3xy' + 3y = 12x^4$ by variation of parameters (lecture only) ESA ★

GIVEN Find the general solution by the method of variation of parameters of $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 3y = 12x^4$ (Cauchy's equation).

STEP 1 $x = e^z$, $z = \log x$: $xy' = Dy$, $x^2y'' = D(D-1)y$ with $D = \frac{d}{dz}$. So $(D^2 - 4D + 3)y = 12e^{4z}$ (leading coefficient 1).

STEP 2 AE $m^2-4m+3=0$, $m=3,1$: $y_c = c_1e^{3z} + c_2e^z$; $y_1 = e^{3z}$, $y_2 = e^z$, $X = 12e^{4z}$.

STEP 3 $W = e^{3z}e^z - e^z\cdot 3e^{3z} = -2e^{4z}$.

STEP 4 $u = -\int \frac{e^z\cdot 12e^{4z}}{-2e^{4z}}dz = 6\int e^z dz = 6e^z$; $v = \int \frac{e^{3z}\cdot 12e^{4z}}{-2e^{4z}}dz = -6\int e^{3z}dz = -2e^{3z}$.

STEP 5 $y_p = 6e^z\cdot e^{3z} - 2e^{3z}\cdot e^z = 4e^{4z} = 4x^4$; $y_c = c_1x^3 + c_2x$.

ANSWER $\boxed{y = c_1x^3 + c_2x + 4x^4}$

CHECK sympy dsolve of the original equation gives $x(C_1 + C_2x^2 + 4x^3)$; $x^2(48x^2) - 3x(16x^3) + 3(4x^4) = 12x^4$. The CF step is from the class 011 Legendre pack.

TRANSCRIPT ONLY T011Vb 00:16:53 · T011Vb 00:18:50 · T011Vb 00:20:49 · T011Vb 00:22:41 · T011Vb 00:23:25

SHORT-NOTE RECAP

$y'' - 2y' - 3y = xe^{-x}$: $W = -4e^{2x}$, $y_p = -\frac{e^{-x}}{16}(x+\frac{1}{4}) - \frac{x^2e^{-x}}{8}$. $(D^2+4)y = \tan 2x$: $W = 2$, $y_p = -\frac{1}{4}\cos 2x\log(\sec 2x + \tan 2x)$ (slide box drops $\cos 2x$). $(D^2-1)y = \frac{2}{1+e^x}$: $W = -2$, $y_p = e^x\log(1+e^{-x}) - e^{-x}\log(1+e^x) - 1$. Cauchy $x^2y'' - 3xy' + 3y = 12x^4$: in z, $W = -2e^{4z}$, $y = c_1x^3 + c_2x + 4x^4$.

3 Trigonometric right-hand sides: $\sec ax$, $\csc x$, $e^x \tan x$, $e^{-x}\sec^3 x$

For complex AE roots $\alpha \pm i\beta$, $y_1 = e^{\alpha x}\cos\beta x$, $y_2 = e^{\alpha x}\sin\beta x$ and $W = \beta e^{2\alpha x}$, so the exponentials cancel whenever $X = e^{\alpha x}g(x)$. What remains are the standard integrals of $\tan$, $\cot$, $\sec$ and $\frac{\sin}{\cos^3}$. Two ESA questions (Jan-24 s1 3a = CW P25, Jan-24 s2 3a) and CW P23, P26, P28 are of this kind.

Formulas

E3.1 · Complex roots $\alpha \pm i\beta$ with $X = e^{\alpha x}g(x)$ ESA ★★★

$$y_1 = e^{\alpha x}\cos\beta x, \quad y_2 = e^{\alpha x}\sin\beta x, \quad W = \beta e^{2\alpha x}$$
$$u = -\frac{1}{\beta}\int g\sin\beta x\,dx, \qquad v = \frac{1}{\beta}\int g\cos\beta x\,dx$$

GIVES The integrals with the exponentials already cancelled.

APPLIES CW P23, P25, P26, P28; ESA Jan-24 s1 3a, s2 3a.

NOTE $\alpha = 0$ for $D^2 + a^2$ ($W = a$); $D^2 - 2D + 2$: $1 \pm i$; $D^2 + 2D + 2$: $-1 \pm i$.

ADDED FOR CLARITY S011V·11 · CW U2 P23 · CW U2 P25 · ESA Jan-24 s1 3a · ESA Jan-24 s2 3a

E3.2 · Trigonometric integrals used ESA ★★★

$$\int \tan ax\,dx = \tfrac{1}{a}\log\sec ax, \qquad \int \cot x\,dx = \log\sin x, \qquad \int \sec x\,dx = \log(\sec x + \tan x)$$
$$\int \frac{\sin^2 x}{\cos x}dx = \log(\sec x + \tan x) - \sin x, \qquad \int \frac{\sin x}{\cos^3 x}dx = \frac{\sec^2 x}{2}$$

GIVES Every integral of Topic 3.

APPLIES After E3.1.

NOTE Strictly $\log|\cdot|$; $\int \frac{\sin x}{\cos^3 x}dx = \frac{\tan^2 x}{2}$ is also right (differs by $\frac{1}{2}$).

S011V·12 · CW U2 P23 · CW U2 P25 · CW U2 P26 · ESA Jan-24 s2 3a

E3.3 · Results: $\sec ax$, $\csc x$, $e^x\tan x$, $e^{-x}\sec^3 x$ ESA ★★★

$$\frac{\sec ax}{D^2+a^2}: \ \frac{\cos ax}{a^2}\log\cos ax + \frac{x}{a}\sin ax, \qquad \frac{\csc x}{D^2+1}: \ -x\cos x + \sin x\log\sin x$$
$$\frac{e^x\tan x}{D^2-2D+2}: \ -e^x\cos x\log(\sec x+\tan x), \qquad \frac{e^{-x}\sec^3 x}{D^2+2D+2}: \ \tfrac{1}{2}e^{-x}\sin x\tan x$$

GIVES The four PIs (W3.1–W3.4).

APPLIES Asked by variation of parameters.

NOTE The last one may also be written $-\frac{e^{-x}\cos 2x}{2\cos x}$ (differs by a CF term).

CW U2 P23 · CW U2 P25 · CW U2 P26 · ESA Jan-24 s1 3a · ESA Jan-24 s2 3a

Graphs and diagrams

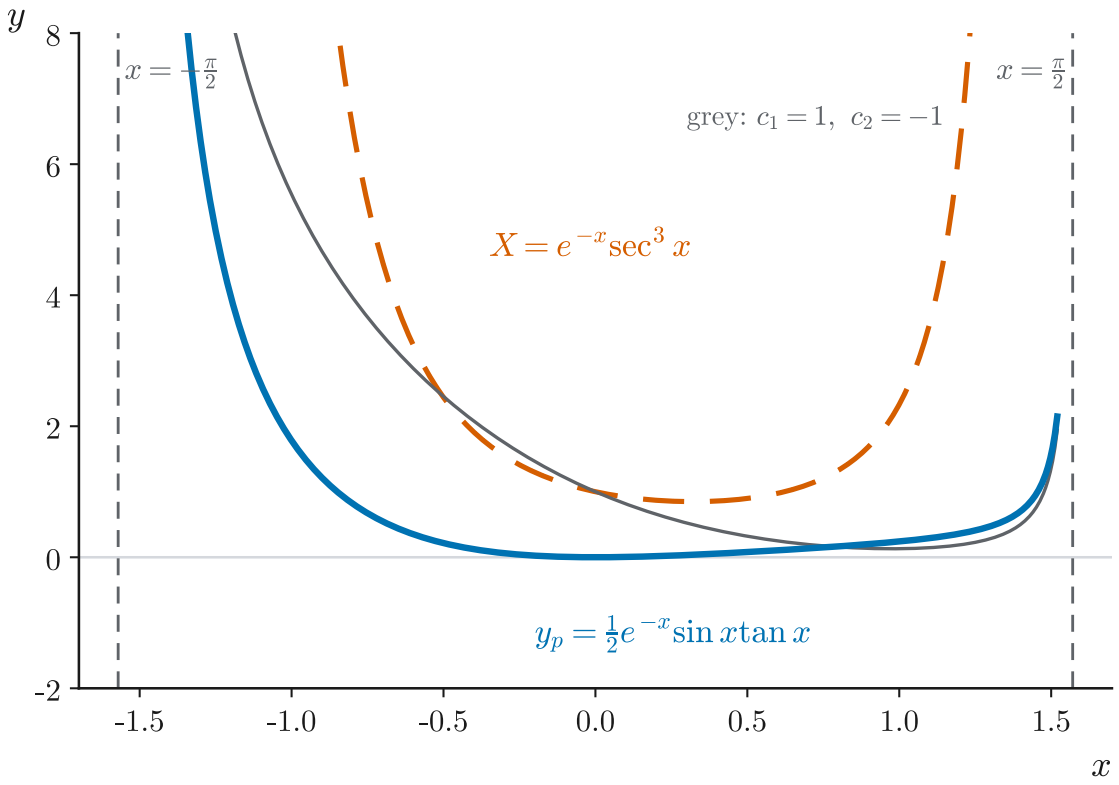
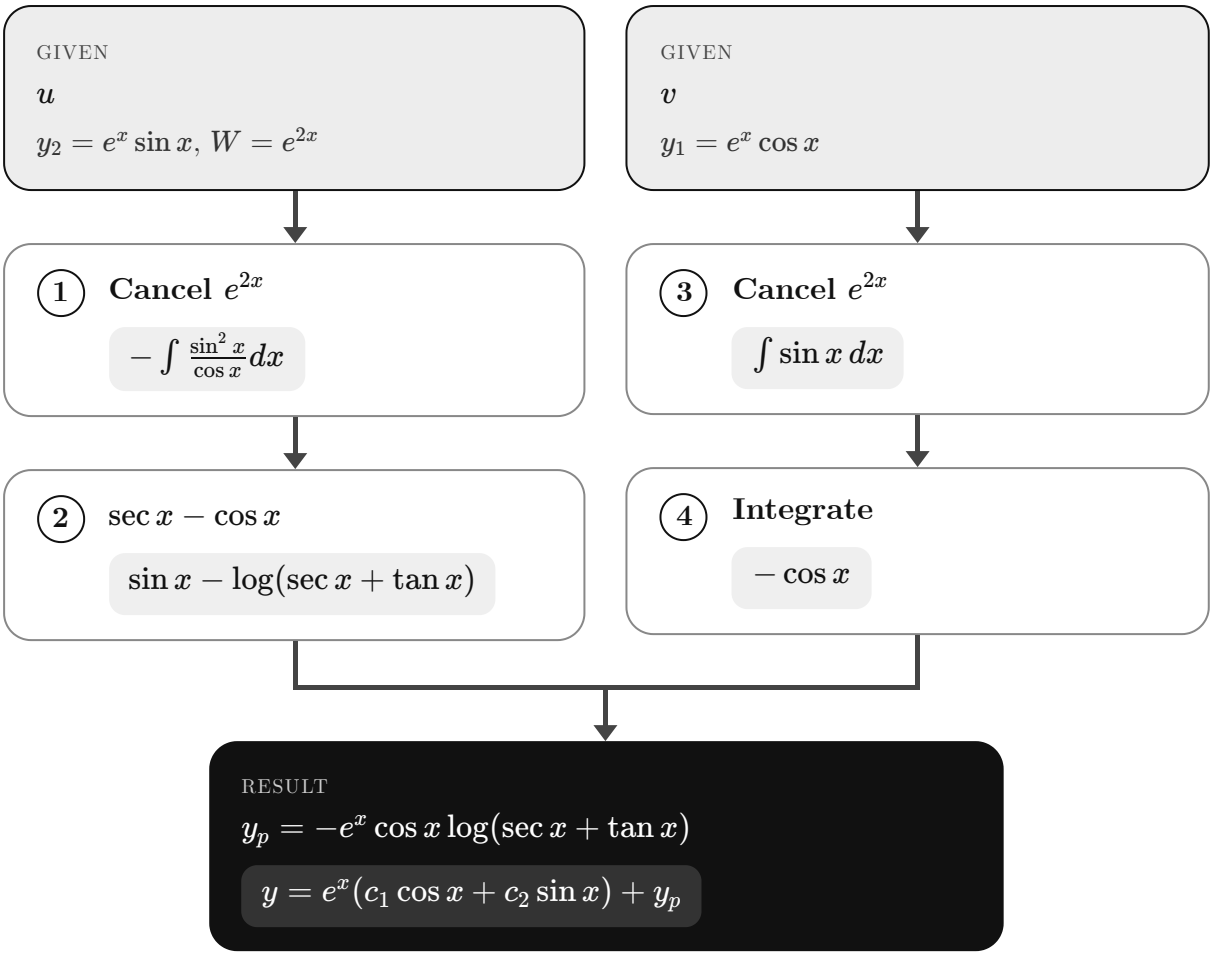


Figure G3.1. ESA Jan-24 s2 3a: $y_p = \frac{1}{2}e^{-x}\sin x \tan x$. ESA ★★★

ADDED FOR CLARITY ESA Jan-24 s2 3a

How to read it. On $-\frac{\pi}{2} < x < \frac{\pi}{2}$, where $\sec x$ exists. Vermilion dashed: $X = e^{-x}\sec^3 x$. Blue: the PI, zero at $x = 0$ and unbounded at the asymptotes $x = \pm\frac{\pi}{2}$ (grey) like X. Grey solid: a general solution with $c_1 = 1$, $c_2 = -1$.

Flowcharts



Remark. The $e^x \sin x \cos x$ terms cancel. Worked in full as W3.3 (7 marks).

Figure F3.1. ESA Jan-24 s1 3a: $(D^2 - 2D + 2)y = e^x \tan x$ ESA ★★★

ESA Jan-24 s1 3a · CW U2 P25

Every variation-of-parameters problem in this pack

Problem	y_1, y_2	W	u	v
xe^{-x}, $D^2 - 2D - 3$ **(W2.1)**	e^{3x}, e^{-x}	$-4e^{2x}$	$-\frac{e^{-4x}}{16}(x + \frac{1}{4})$	$-\frac{x^2}{8}$
$\tan 2x$, $D^2 + 4$ **(W2.2)**	$\cos 2x$, $\sin 2x$	2	$\frac{1}{4}[\sin 2x - \log(\sec 2x + \tan 2x)]$	$-\frac{\cos 2x}{4}$
$\frac{2}{1+e^x}$, $D^2 - 1$ **(W2.3)**	e^x, e^{-x}	-2	$-e^{-x} + \log(1 + e^{-x})$	$-\log(1 + e^x)$
$\sec ax$ **(W3.1)**	$\cos ax$, $\sin ax$	a	$\frac{1}{a^2} \log \cos ax$	$\frac{x}{a}$

Problem	y_1, y_2	W	u	v
$\csc x$ **(W3.2)**	$\cos x$, $\sin x$	1	$-x$	$\log \sin x$
$e^x \tan x$ **(W3.3, ESA)**	$e^x \cos x$, $e^x \sin x$	e^{2x}	$\sin x - \log(\sec x + \tan x)$	$-\cos x$
$e^{-x} \sec^3 x$ **(W3.4, ESA)**	$e^{-x} \cos x$, $e^{-x} \sin x$	e^{-2x}	$-\frac{\sec^2 x}{2}$	$\tan x$
$\frac{e^{-3x}}{x}$ **(W4.1, ESA)**	e^{-3x}, xe^{-3x}	e^{-6x}	$-x$	$\log x$
$\frac{e^{x}}{x}$ **(W4.2, ESA)**	e^{x}, xe^{x}	e^{2x}	$-x$	$\log x$
$\frac{e^{-x}}{x^2}$ **(W4.3)**	e^{-x}, xe^{-x}	e^{-2x}	$-\log x$	$-\frac{1}{x}$
$\frac{25}{2}e^{-x}$, $-3y$ **(W4.4)**	$e^{3x/2}$, e^{-x}	$-\frac{5}{2}e^{x/2}$	$-2e^{-5x/2}$	$-5x$

S011V·7 · S011V·11 · S011V·14 · CW U2 P23 · CW U2 P24 · CW U2 P25 · CW U2 P26 · CW U2 P27 · ESA Jul-23 3c · ESA May-24 3c · ESA Jan-24 s2 3a

Important points and why

1. With complex roots and $X = e^{\alpha x} g$, the exponentials cancel: $W = \beta e^{2\alpha x}$ (E3.1). ESA Jan-24 s1 3a and s2 3a are this pattern. ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s1 3a · ESA Jan-24 s2 3a · CW U2 P25
2. Know the integrals: $\int \tan = \log \sec$, $\int \cot = \log \sin$, $\int \sec = \log(\sec + \tan)$, $\int \frac{\sin}{\cos^3} = \frac{\sec^2}{2}$. ESA ★★★ CW U2 P23 · CW U2 P26 · ESA Jan-24 s2 3a · S011V·12

Worked examples

W3.1 · $y'' + a^2 y = \sec ax$ (CW P23) — ESA ★★

GIVEN Solve $\frac{d^2y}{dx^2} + a^2 y = \sec ax$ by the method of variation of parameters.

STEP 1 AE $m = \pm ia$: $y_1 = \cos ax$, $y_2 = \sin ax$; $W = \cos ax \cdot a\cos ax + \sin ax \cdot a \sin ax = a$.

STEP 2 $u = -\int \frac{\sin ax \sec ax}{a} dx = -\frac{1}{a}\int \tan ax\, dx = -\frac{1}{a^2}\log \sec ax = \frac{1}{a^2}\log \cos ax$.

STEP 3 $v = \int \frac{\cos ax \sec ax}{a} dx = \frac{1}{a}\int dx = \frac{x}{a}$.

STEP 4 $y_p = \frac{\cos ax}{a^2}\log \cos ax + \frac{x}{a}\sin ax$.

ANSWER $$\boxed{y = C_1 \cos ax + C_2 \sin ax + \frac{\cos ax}{a^2}\log \cos ax + \frac{x}{a}\sin ax}$$

CHECK CW answer agrees. sympy (symbolic a): $W = a$, u, v as above, residual 0. ($\log|\cos ax|$ strictly.)

CW U2 P23

W3.2 · $y'' + y = \csc x$ (CW P26) ESA ★★

GIVEN Solve $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$ by the method of variation of parameters.

STEP 1 $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$.

STEP 2 $u = -\int \sin x \csc x \, dx = -x$; $v = \int \cos x \csc x \, dx = \int \cot x \, dx = \log \sin x$.

STEP 3 $y_p = -x \cos x + \sin x \log \sin x$.

ANSWER $\boxed{y = C_1 \cos x + C_2 \sin x - x \cos x + \sin x \log \sin x}$

CHECK CW answer agrees; sympy residual 0.

CW U2 P26

W3.3 · $(D^2 - 2D + 2)y = e^x \tan x$ (ESA Jan-24 s1 3a = CW P25) ESA ★★★

GIVEN Solve $(D^2 - 2D + 2)y = e^x \tan x$ by the method of variation of parameters (7 marks).

STEP 1 AE $m^2 - 2m + 2 = 0$, $m = 1 \pm i$: $y_c = e^x(c_1 \cos x + c_2 \sin x)$; $y_1 = e^x \cos x$, $y_2 = e^x \sin x$.

STEP 2 $W = e^x \cos x \cdot e^x(\sin x + \cos x) - e^x \sin x \cdot e^x(\cos x - \sin x) = e^{2x}$.

STEP 3 $u = -\int \frac{e^x \sin x \cdot e^x \tan x}{e^{2x}} dx = -\int \frac{\sin^2 x}{\cos x} dx = -\int (\sec x - \cos x) dx = \sin x - \log(\sec x + \tan x)$.

STEP 4 $v = \int \frac{e^x \cos x \cdot e^x \tan x}{e^{2x}} dx = \int \sin x \, dx = -\cos x$.

STEP 5 $y_p = e^x \cos x[\sin x - \log(\sec x + \tan x)] - e^x \sin x \cos x = -e^x \cos x \log(\sec x + \tan x)$.

ANSWER $\boxed{y = e^x(c_1 \cos x + c_2 \sin x) - e^x \cos x \, \log(\sec x + \tan x)}$

CHECK sympy: $W = e^{2x}$, u', v' match, residual 0. CW P25's printed answer reads "$e^x \cos \log(\sec x + \tan x)$": the x of $\cos x$ is missing. Figure F3.1.

ESA Jan-24 s1 3a · CW U2 P25

W3.4 · $y''+2y'+2y=e^{-x}\sec^3 x$ (ESA Jan-24 s2 3a) ESA ★★★

GIVEN Solve $y''+2y'+2y=e^{-x}\sec^3 x$ by the method of variation of parameters (7 marks).

STEP 1 AE $m^2+2m+2=0$, $m=-1\pm i$: $y_c=e^{-x}(c_1\cos x+c_2\sin x)$; $y_1=e^{-x}\cos x$, $y_2=e^{-x}\sin x$.

STEP 2 $W=e^{-2x}$ (E3.1 with $\alpha=-1$, $\beta=1$).

STEP 3 $u=-\int\frac{e^{-x}\sin x\cdot e^{-x}\sec^3 x}{e^{-2x}}dx=-\int\frac{\sin x}{\cos^3 x}dx=-\frac{1}{2\cos^2 x}$ ($t=\cos x$).

STEP 4 $v=\int\frac{e^{-x}\cos x\cdot e^{-x}\sec^3 x}{e^{-2x}}dx=\int\sec^2 x\,dx=\tan x$.

STEP 5 $y_p=-\frac{e^{-x}\cos x}{2\cos^2 x}+\frac{e^{-x}\sin^2 x}{\cos x}=e^{-x}\frac{2\sin^2 x-1}{2\cos x}=-\frac{e^{-x}\cos 2x}{2\cos x}$.

STEP 6 Equivalently, $-\frac{e^{-x}\cos 2x}{2\cos x}=\frac{e^{-x}\sin^2 x}{2\cos x}-\frac{1}{2}e^{-x}\cos x$, and $-\frac{1}{2}e^{-x}\cos x$ is absorbed into c_1.

ANSWER $$\boxed{y=e^{-x}(c_1\cos x+c_2\sin x)+\frac{1}{2}e^{-x}\sin x\tan x}$$

CHECK sympy: residual 0 for both forms; dsolve gives $e^{-x}(C_1\sin x+C_2\cos x-\frac{\cos 2x}{2\cos x})$. Using $u=-\frac{\tan^2 x}{2}$ gives the $\frac{1}{2}e^{-x}\sin x\tan x$ form directly. Figure G3.1.

ESA Jan-24 s2 3a

Past-paper questions

ESA Jan 2024 set 1 · 7 marks · UE23MA141A Q3.a · solved in W3.3

Solve $(D^2-2D+2)y=e^x\tan x$ by the method of variation of parameters (= CW P25).

ESA Jan 2024 set 2 · 7 marks · UE23MA141A Q3.a · solved in W3.4

Solve $y''+2y'+2y=e^{-x}\sec^3 x$ by the method of variation of parameters.

SHORT-NOTE RECAP

Roots $\alpha\pm i\beta$, $X=e^{\alpha x}g$: $W=\beta e^{2\alpha x}$, $u=-\frac{1}{\beta}\int g\sin\beta x$, $v=\frac{1}{\beta}\int g\cos\beta x$. $\sec ax$: $\frac{\cos ax}{a^2}\log\cos ax+\frac{x}{a}\sin ax$. $\csc x$: $-x\cos x+\sin x\log\sin x$. $e^x\tan x$ (ESA): $-e^x\cos x\log(\sec x+\tan x)$. $e^{-x}\sec^3 x$ (ESA): $\frac{1}{2}e^{-x}\sin x\tan x$.

4 Repeated roots with $X = \frac{e^{mx}}{x^k}$, and the leading-coefficient trap

For a repeated root m, $y_1 = e^{mx}$, $y_2 = xe^{mx}$ and $W = e^{2mx}$; with $X = e^{mx}g(x)$ the integrals reduce to $u = -\int xg\,dx$, $v = \int g\,dx$. With $g = \frac{1}{x}$ (ESA Jul-23 3c = Jun-24 3a, May-24 3c) the PI is $xe^{mx}\log x$ after absorbing $-xe^{mx}$ into the CF; with $g = \frac{1}{x^2}$ (CW P24) it is $-e^{mx}\log x$. CW P27 has leading coefficient 2: divide by 2 before using X.

Formulas

E4.1 · Repeated root m with $X = e^{mx}g(x)$ ESA ★★★

$$y_1 = e^{mx}, \quad y_2 = xe^{mx}, \quad W = e^{2mx}, \qquad u = -\int x\,g\,dx, \quad v = \int g\,dx$$

GIVES The integrals with the exponentials cancelled.

APPLIES $(D-m)^2y = e^{mx}g(x)$: ESA Jul-23 3c, May-24 3c, CW P24.

NOTE $g = \frac{1}{x}$: $u = -x$, $v = \log x$. $g = \frac{1}{x^2}$: $u = -\log x$, $v = -\frac{1}{x}$.

ADDED FOR CLARITY ESA Jul-23 3c · ESA May-24 3c · CW U2 P24

E4.2 · Results: $\frac{e^{mx}}{x}$ and $\frac{e^{mx}}{x^2}$ ESA ★★★

$$(D-m)^2y = \frac{e^{mx}}{x}: \quad y = (c_1 + c_2x)e^{mx} + xe^{mx}\log x$$

$$(D-m)^2y = \frac{e^{mx}}{x^2}: \quad y_p = -e^{mx}\log x$$

GIVES The ESA answers with $m = -3$ (Jul-23, Jun-24) and $m = 1$ (May-24); CW P24 with $m = -1$.

APPLIES After absorbing CF terms.

NOTE Raw $uy_1 + vy_2$ has an extra $-xe^{mx}$ (resp. $-e^{mx}$): a CF term.

ESA Jul-23 3c · ESA Jun-24 3a · ESA May-24 3c · CW U2 P24

Graphs and diagrams

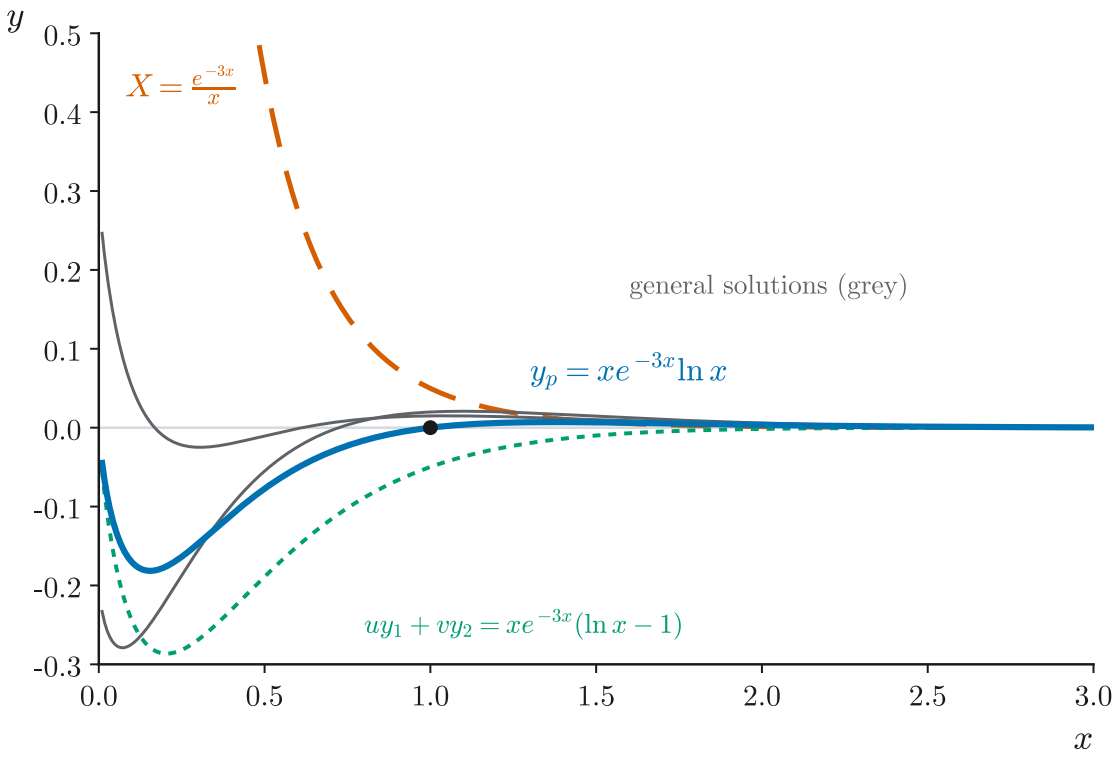
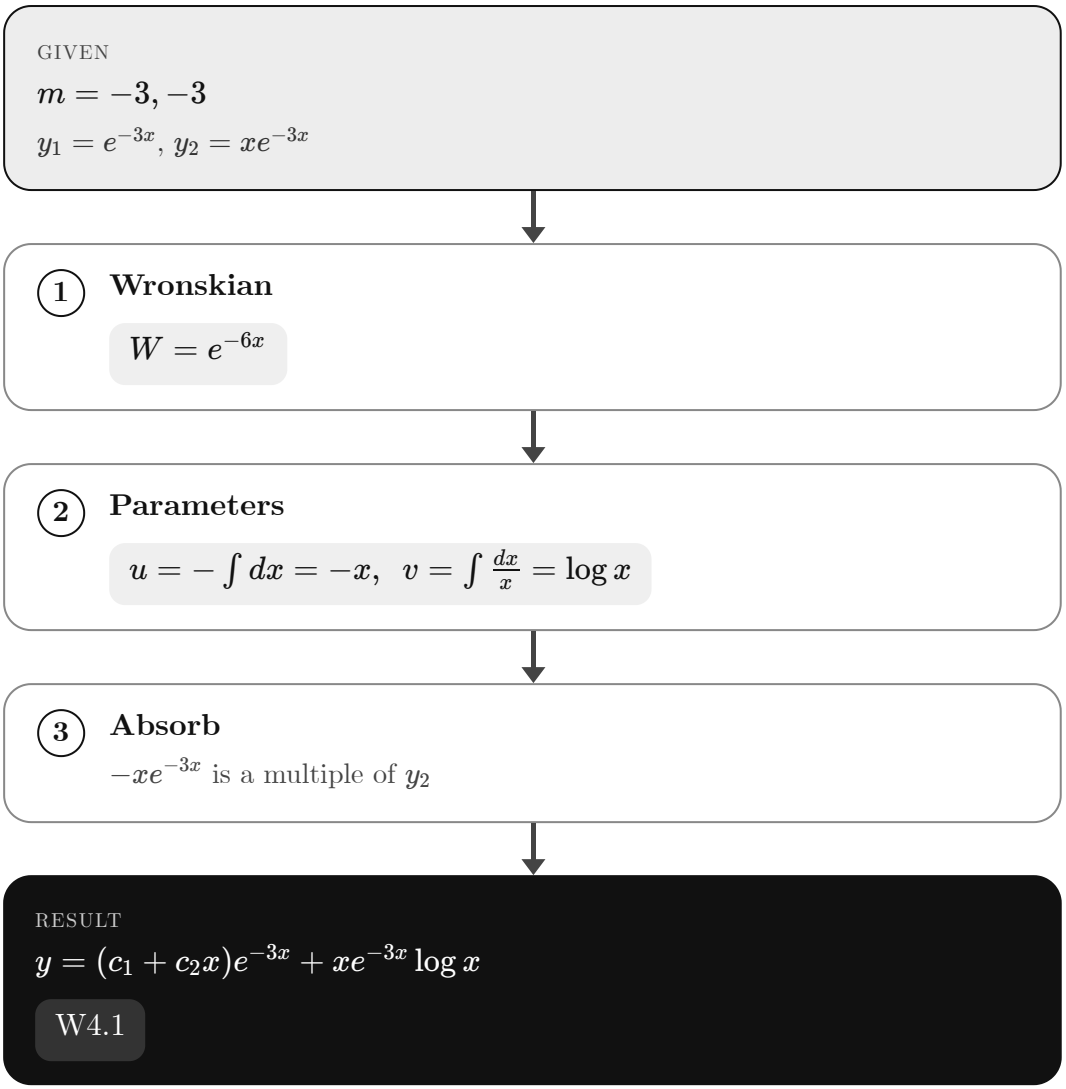


Figure G4.1. ESA Jul-23 3c: $y_p = xe^{-3x}\log x$ and general solutions. ESA ★★★

ADDED FOR CLARITY ESA Jul-23 3c · ESA Jun-24 3a

How to read it. For $x > 0$. Vermilion dashed: $X = \frac{e^{-3x}}{x}$, unbounded at $x = 0$. Blue: $y_p = xe^{-3x}\log x$, negative for $x < 1$, zero at $x = 1$, tending to 0. Grey: two general solutions $(c_1 + c_2x)e^{-3x} + y_p$; the dotted green curve is the un-absorbed $uy_1 + vy_2 = xe^{-3x}(\log x - 1)$, itself one of the general solutions ($c_2 = -1$).

Flowcharts



Remark. May-24 3c is the same with $m = 1$: $y = (c_1 + c_2 x)e^{x} + xe^{x}\log x$.

Figure F4.1. ESA Jul-23 3c: $y'' + 6y' + 9y = \frac{e^{-3x}}{x}$ ESA ★★★

ESA Jul-23 3c · ESA Jun-24 3a

Terms you may absorb into the CF

Problem	Raw $uy_1 + vy_2$	Absorbed term	Final PI
W2.1	$-\frac{x^2e^{-x}}{8} - \frac{xe^{-x}}{16} - \frac{e^{-x}}{64}$	$-\frac{e^{-x}}{64}$ (into c_2)	$-\frac{x^2e^{-x}}{8} - \frac{xe^{-x}}{16}$
W4.1	$xe^{-3x}\log x - xe^{-3x}$	$-xe^{-3x}$	$xe^{-3x}\log x$
W4.2	$xe^{x}\log x - xe^{x}$	$-xe^{x}$	$xe^{x}\log x$
W4.3	$-e^{-x}\log x - e^{-x}$	$-e^{-x}$	$-e^{-x}\log x$
W4.4	$-2e^{-x} - 5xe^{-x}$	$-2e^{-x}$ (optional)	$-5xe^{-x}$
W3.4	$-\frac{e^{-x}\cos 2x}{2\cos x}$	$-\frac{1}{2}e^{-x}\cos x$	$\frac{1}{2}e^{-x}\sin x\tan x$
W2.3	$e^{x}\log(1+e^{-x}) - e^{-x}\log(1+e^{x}) - 1$	none (-1 is not in the CF)	unchanged

ADDED FOR CLARITY S011V·10 · ESA Jul-23 3c · ESA May-24 3c · CW U2 P24 · CW U2 P27 · ESA Jan-24 s2 3a

Important points and why

1. ESA Jul-23 3c = Jun-24 3a and May-24 3c are the same pattern: $(D-m)^2y = \frac{e^{mx}}{x}$, answer $y = (c_1 + c_2x)e^{mx} + xe^{mx}\log x$. ESA ★★★ ESA Jul-23 3c · ESA Jun-24 3a · ESA May-24 3c
2. State it when you absorb a term: "$-xe^{mx}$ is a multiple of y_2 and is absorbed into c_2". ESA ★★
   ADDED FOR CLARITY ESA Jul-23 3c · CW U2 P24

Why can the operator rules not do $\frac{e^{-3x}}{x}$?
Type 4 gives $e^{-3x}\frac{1}{D^2}\frac{1}{x}$, i.e. integrating $\frac{1}{x}$ twice, which is possible ($x\log x - x$) — the VoP route just organises the same integrals. The question asks for VoP, so use it.
ADDED FOR CLARITY ESA Jul-23 3c

Exam trap. CW P27 ($2y'' - y' - \cdots = 25e^{-x}$): divide by 2 first, so $X = \frac{25}{2}e^{-x}$, not $25e^{-x}$. CW U2 P27 · S011V·6

Exam trap. "Solve" means $y = y_c + y_p$: the ESA questions want the general solution, not only y_p.
ADDED FOR CLARITY ESA Jul-23 3c · ESA May-24 3c

Worked examples

W4.1 · $y'' + 6y' + 9y = \frac{e^{-3x}}{x}$ (ESA Jul-23 3c = Jun-24 3a) ESA ★★★

GIVEN Solve $y'' + 6y' + 9y = \frac{e^{-3x}}{x}$ by the method of variation of parameters (6 marks).

STEP 1 AE $(m+3)^2 = 0$, $m = -3, -3$: $y_c = (c_1 + c_2 x)e^{-3x}$; $y_1 = e^{-3x}$, $y_2 = xe^{-3x}$.

STEP 2 $W = e^{-3x}(1-3x)e^{-3x} - xe^{-3x}(-3e^{-3x}) = e^{-6x}$.

STEP 3 $u = -\int \frac{xe^{-3x}\cdot e^{-3x}/x}{e^{-6x}}dx = -\int dx = -x$; $v = \int \frac{e^{-3x}\cdot e^{-3x}/x}{e^{-6x}}dx = \int \frac{dx}{x} = \log x$.

STEP 4 $y_p = -xe^{-3x} + xe^{-3x}\log x$; $-xe^{-3x}$ is a multiple of y_2: absorbed into c_2.

ANSWER $\boxed{y = (c_1 + c_2 x)e^{-3x} + xe^{-3x}\log x}$

CHECK sympy: residual 0; dsolve gives $(C_1 + x(C_2 + \log x))e^{-3x}$, the same. Figure G4.1, Figure F4.1.

ESA Jul-23 3c · ESA Jun-24 3a

W4.2 · $y'' - 2y' + y = \frac{e^x}{x}$ (ESA May-24 3c) ESA ★★★

GIVEN Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = \frac{e^x}{x}$ by the method of variation of parameters (6 marks).

STEP 1 AE $(m-1)^2 = 0$, $m = 1, 1$: $y_c = (c_1 + c_2 x)e^x$; $y_1 = e^x$, $y_2 = xe^x$; $W = e^x(1+x)e^x - xe^x \cdot e^x = e^{2x}$.

STEP 2 $u = -\int \frac{xe^x\cdot e^x/x}{e^{2x}}dx = -x$; $v = \int \frac{e^x\cdot e^x/x}{e^{2x}}dx = \log x$.

STEP 3 $y_p = -xe^x + xe^x\log x$; absorb $-xe^x$ into $c_2 xe^x$.

ANSWER $\boxed{y = (c_1 + c_2 x)e^x + xe^x\log x}$

CHECK sympy: residual 0. Same pattern as W4.1 with $m = 1$ (E4.2).

ESA May-24 3c

W4.3 · $y'' + 2y' + y = \frac{e^{-x}}{x^2}$ (CW P24) ESA ★★

GIVEN Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = \frac{e^{-x}}{x^2}$ by the method of variation of parameters.

STEP 1 $m = -1, -1$: $y_1 = e^{-x}$, $y_2 = xe^{-x}$, $W = e^{-2x}$.

STEP 2 $u = -\int \frac{xe^{-x}\cdot e^{-x}/x^2}{e^{-2x}}dx = -\int \frac{dx}{x} = -\log x$; $v = \int \frac{e^{-x}\cdot e^{-x}/x^2}{e^{-2x}}dx = \int \frac{dx}{x^2} = -\frac{1}{x}$.

STEP 3 $y_p = -e^{-x}\log x - \frac{1}{x}\cdot xe^{-x} = -e^{-x}\log x - e^{-x}$; absorb $-e^{-x}$ into C_1.

ANSWER $\boxed{y = C_1 e^{-x} + C_2 xe^{-x} - e^{-x}\log x}$

CHECK CW answer agrees after the absorption (the sheet does not mention it). sympy residual 0.

CW U2 P24

W4.4 · $2y'' - y' - 3y = 25e^{-x}$ (CW P27; divide by 2 first) ESA ★★

GIVEN CW P27, printed "$2\frac{d^2y}{dx^2} - \frac{dy}{dx} - y + = 25e^{-x}$"; its answer fits $2y'' - y' - 3y = 25e^{-x}$. Solve by variation of parameters.

STEP 1 Standard form: $y'' - \frac{1}{2}y' - \frac{3}{2}y = \frac{25}{2}e^{-x}$, so $X = \frac{25}{2}e^{-x}$.

STEP 2 AE $2m^2 - m - 3 = 0$, $(2m-3)(m+1) = 0$, $m = \frac{3}{2}, -1$: $y_1 = e^{3x/2}$, $y_2 = e^{-x}$; $W = e^{3x/2}(-e^{-x}) - e^{-x} \cdot \frac{3}{2}e^{3x/2} = -\frac{5}{2}e^{x/2}$.

STEP 3 $u = -\int \frac{e^{-x} \cdot \frac{25}{2}e^{-x}}{-\frac{5}{2}e^{x/2}}dx = 5\int e^{-5x/2}dx = -2e^{-5x/2}$; $v = \int \frac{e^{3x/2} \cdot \frac{25}{2}e^{-x}}{-\frac{5}{2}e^{x/2}}dx = -5\int dx = -5x$.

STEP 4 $y_p = -2e^{-x} - 5xe^{-x}$ ($-2e^{-x}$ could be absorbed into C_2e^{-x}).

STEP 5 As printed with $-y$: AE $2m^2 - m - 1 = 0$, $m = 1, -\frac{1}{2}$, $W = -\frac{3}{2}e^{x/2}$, $u = -\frac{25}{6}e^{-2x}$, $v = \frac{50}{3}e^{-x/2}$, $y_p = \frac{25}{2}e^{-x}$.

ANSWER $y = C_1e^{3x/2} + C_2e^{-x} - 2e^{-x} - 5xe^{-x}$ (with $-3y$, the CW answer); with $-y$: $y = C_1e^{x} + C_2e^{-x/2} + \frac{25}{2}e^{-x}$

CHECK sympy: both residuals 0; dsolve gives $C_2e^{3x/2} + (C_1 - 5x)e^{-x}$ and $C_1e^{-x/2} + C_2e^{x} + \frac{25}{2}e^{-x}$. Without dividing by 2 every u, v would be twice too large.

CW U2 P27 · S011V·6

Past-paper questions

ESA Jul 2023 (6 marks) UE22MA141A Q3.c · solved in W4.1

Solve $y'' + 6y' + 9y = \frac{e^{-3x}}{x}$ by the method of variation of parameters.

ESA Jun 2024 (6 marks) UE23MA141A Q3.a · solved in W4.1

Solve $y'' + 6y' + 9y = \frac{e^{-3x}}{x}$ by the method of variation of parameters (repeat of Jul-23 3c).

ESA May 2024 (6 marks) UE22MA141A Q3.c · solved in W4.2

Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = \frac{e^x}{x}$ by the method of variation of parameters.

SHORT-NOTE RECAP

Repeated root m: $y_1 = e^{mx}$, $y_2 = xe^{mx}$, $W = e^{2mx}$. $X = \frac{e^{mx}}{x}$ (ESA ×3): $u = -x$, $v = \log x$, $y = (c_1 + c_2x)e^{mx} + xe^{mx}\log x$. $X = \frac{e^{-x}}{x^2}$: $y_p = -e^{-x}\log x$. Leading coefficient 2: divide first (CW P27).

A Symbol glossary

Symbol	Name	Meaning	Appears in
X	Right-hand side	Function of x on the right of the standard form $y'' + py' + qy = X$ (capital)	E1.1, E1.3
p, q	Coefficients	Constants of y', y after dividing by the coefficient of y''	E1.1
y_1, y_2	CF solutions	Linearly independent solutions of the homogeneous equation; $y_c = c_1 y_1 + c_2 y_2$	E1.1, E1.2
W	Wronskian	$y_1 y_2' - y_2 y_1'$; never zero here	E1.2, E3.1, E4.1
u, v	Varying parameters	$u = -\int \frac{y_2 X}{W} dx$, $v = \int \frac{y_1 X}{W} dx$; functions of x (some books: A, B)	E1.3
y_c, y_p	CF and PI	Complementary function and particular integral; $y = y_c + y_p$	E1.4
c_1, c_2	Arbitrary constants	In the CF only; absorb constant multiples of y_1, y_2	E1.4
$\alpha \pm i\beta$	Complex AE roots	$y_1 = e^{\alpha x}\cos\beta x$, $y_2 = e^{\alpha x}\sin\beta x$	E3.1
m	AE root	Repeated root in Topic 4 ($y_2 = xe^{mx}$)	E4.1, E4.2
$g(x)$	Reduced right-hand side	$X = e^{\alpha x} g$ or $e^{mx} g$; what is left after the exponentials cancel	E3.1, E4.1
z, D	Cauchy variable	$z = \log x$, $D = \frac{d}{dz}$ in W2.4; elsewhere $D = \frac{d}{dx}$ (clash)	W2.4
t	Substitution variable	$t = 1 + e^{-x}$ or $t = \cos x$ inside an integral (not time)	E2.1
$\log$	Natural logarithm	$\log_e$; strictly $\log\lvert\cdot\rvert$ in $\log\cos ax$, $\log\sin x$, $\log x$	E3.2, E4.2

B Sources, coverage and gaps

Source	Path	Use
Deck 011 VoP (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.*	18 slides; content on 4–17 (every equation read from the rendered images; .slides.md has titles only)
Transcript T011V (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.*	14.9 min; slides 4–10 (method, example xe^{-x})
Transcript T011Vb (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method_Continued__97963caf81f64a9c.*	24.5 min; slides 11–17 ($\tan 2x$, $\frac{2}{1+e^x}$) and the lecture-only Cauchy example
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 23–28 ("Solve by method of variation of parameters only"), page 5–6 rendered and every answer recomputed
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	Jul-23 3c, Jun-24 3a, May-24 3c, Jan-24 s1 3a, Jan-24 s2 3a, via the shared ESA map
Video 011 VoP (.mp4, both parts)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method*.mp4	Not opened: every slide equation is readable and no video trigger applies

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences). Variation of parameters is asked in 4 distinct questions, 5 occurrences, every time worth 6–7 marks and every time with the words "by the method of variation of parameters": Jul-23 3c = Jun-24 3a ($y''+6y'+9y=\frac{e^{-3x}}{x}$), May-24 3c ($y''-2y'+y=\frac{e^x}{x}$), Jan-24 s1 3a ($(D^2-2D+2)y=e^x\tan x$, = CW P25) and Jan-24 s2 3a ($y''+2y'+2y=e^{-x}\sec^3 x$). It appears in 5 of the 8 papers, so the method, the Wronskian and the u, v formulas are ★★★; the $\frac{e^{mx}}{x}$ repeated-root pattern (2 distinct, 3 occurrences) and $e^{\alpha x}\times$trig ($\tan$, $\sec^3$) are ★★★; the slide-only right-hand sides (xe^{-x}, $\frac{2}{1+e^x}$) and the CW-only ones ($\sec ax$, $\csc x$, $\tan 2x$) are ★★; the use on variable-coefficient equations (lecture only, never asked) is ★.

Known gaps, conflicts and corrections

1. Slide 011V·13's boxed answer "Thus, $y_p=-\frac{1}{4}\log(\sec 2x+\tan 2x)$" drops the factor $\cos 2x$. The line above it on the same slide gives $\frac{1}{4}[\sin 2x-\log(\sec 2x+\tan 2x)]\cos 2x+\sin 2x\cdot\left(-\frac{\cos 2x}{4}\right)=-\frac{1}{4}\cos 2x\log(\sec 2x+\tan 2x)$, which is CW P28's answer; substituting the boxed expression back does not give $\tan 2x$ (sympy). W2.2 and a trap.
2. T011Vb 00:06:49 says "v is equal to minus $\cos 2x$ by 2"; the slide (011V·13) and the working give $v=-\frac{\cos 2x}{4}$. T011Vb 00:15:56 reads u as "minus e power x plus log ..."; it is $-e^{-x}+\log(1+e^{-x})$ as on slide 011V·16.

3. Every other slide line was recomputed with sympy: 011V·8 ($W = -4e^{2x}$), 011V·9 ($u = -\frac{e^{-4x}}{16}(x + \frac{1}{4})$), 011V·10 ($v = -\frac{x^2}{8}$; PI $-\frac{e^{-x}}{16}(x + \frac{1}{4}) - \frac{x^2 e^{-x}}{8}$), 011V·11 ($W = 2$), 011V·12 ($u$), 011V·13 ($v$), 011V·14 ($W = -2$), 011V·15–16 ($u = -e^{-x} + \log(1 + e^{-x})$) and 011V·17 ($v = -\log(1 + e^{x})$; $y_p = e^{x}\log(1 + e^{-x}) - e^{-x}\log(1 + e^{x}) - 1$) are right. The $-\frac{e^{-x}}{64}$ in 011V·10's PI is a multiple of $y_2 = e^{-x}$ and can be absorbed into c_2; the Type 4 rule (class 006) gives $-\frac{x^2 e^{-x}}{8} - \frac{x e^{-x}}{16}$, the same up to that term.

4. Slides 011V·7–10 and 011V·11–13 ask only for the particular integral; 011V·14 asks for the general solution but ends with "Thus, $y = y_c + y_p$". The full general solutions are written out here.

5. CW P27 is printed "$2\frac{d^2y}{dx^2} - \frac{dy}{dx} - y + = 25e^{-x}$" (a character is lost before "="). Its printed answer $C_1 e^{\frac{3}{2}x} + C_2 e^{-x} - 2e^{-x} - 5xe^{-x}$ is the solution of $2y'' - y' - 3y = 25e^{-x}$ (AE roots $\frac{3}{2}, -1$), not of $2y'' - y' - y = 25e^{-x}$ (roots $1, -\frac{1}{2}$; $y = C_1 e^{x} + C_2 e^{-\frac{x}{2}} + \frac{25}{2}e^{-x}$). Both are solved in W4.4 (sympy-checked); with $-3y$ the term $-2e^{-x}$ could be absorbed into $C_2 e^{-x}$, so the printed answer keeps a removable term.

6. CW P25's answer is printed "$-e^{x}\ \cos\log(\sec x + \tan x)$": the x of $\cos x$ is missing; it is $-e^{x}\cos x \log(\sec x + \tan x)$. CW P24's answer omits the term $v\,y_2 = -e^{-x}$ (absorbed into $C_1 e^{-x}$). CW P23, P26, P28 answers are right as printed.

7. ESA Jan-24 s2 3a: depending on whether $\int \frac{\sin x}{\cos^3 x}dx$ is written $\frac{\sec^2 x}{2}$ or $\frac{\tan^2 x}{2}$, the PI comes out as $-\frac{e^{-x}\cos 2x}{2\cos x}$ or $\frac{1}{2}e^{-x}\sin x \tan x$; they differ by $-\frac{1}{2}e^{-x}\cos x$, a CF term, so both are correct.

8. The Cauchy equation $x^2y'' - 3xy' + 3y = 12x^4$ (T011Vb 00:16:53–00:23:47) is on no slide (transcript only). Its CF needs $z = \log x$, from the class 011 Legendre pack; the variation-of-parameters part is done in z.

9. T011V 00:02:06 says y_1, y_2 are linearly independent "if the linear combination $c_1y_1 + c_2y_2$ is not equal to 0"; the precise statement is that $c_1y_1 + c_2y_2 \equiv 0$ only for $c_1 = c_2 = 0$ (class 001). Corrected as a trap, flagged added for clarity.

10. The deck states the method for constant p, q and says it also works for variable coefficients if the CF is known; T011V 00:05:39 adds that it extends to n-th order equations. Only second order is taught or asked; the derivation of u, v (E1.5) is added for clarity.

11. Transcript recognition errors were corrected silently: "Ronskeian", "Ronskean", "Ronskiana", "Rand scheme", "RAN scale", "Ron scale", "Rans scan", "Ran scale", "Roll scan", "ron skin", "runs here", "long scale", "wrong scale", "r scale" → Wronskian; "by 1", "by 2", "I1", "byte" → y_1, y_2; "u and b", "b is" → u and v; "variation or parameters", "my parameters" → variation of parameters; "LTE", "LBE", "differential equity" → LDE; "wtx", "ts", "DS", "d is dz" → dx, dz; "psi 2x", "sine to x", "cos to x", "stand 2x", "car square 2x", "second 2x" → $\sin 2x$, $\cos 2x$, $\tan 2x$, $\cos^2 2x$, $\sec 2x$; "c po x", "E po x", "e bar" → e^{x}, e power; "log p" → $\log t$; "observed" → absorbed; "D by dz" in T011V 00:07:48 (self-corrected to $\frac{d}{dx}$); "Said" → z; "integrality valid" → integral to evaluate.

Coverage

Source	Citations	Slides cited (slide × citations)
S011V	77	4×6, 5×13, 6×10, 7×4, 8×4, 9×3, 10×5, 11×3, 12×6, 13×4, 14×4, 15×4, 16×4, 17×7
CW	36	
ESA	45	
T011V	40	
T011Vb	30	

The deck's own title slide says “SESSION: 8”; the bucket numbers it 011, and the bucket puts both this deck and the Legendre deck (“SESSION: 9”) under 011 — hence the tags S011V and S011L (bucket numbers are not contiguous: the decks say SESSION 1–11; 007, 008, 010, 012 and 013 do not exist; 009 = session 7, 011 = sessions 8 and 9, 014 = 10, 015 = 11). Slides 1–3 and 18 are title, class-content and closing slides; slides 4–6 the method; 7–10 example xe^{-x}; 11–13 $\tan 2x$; 14–17 $\frac{2}{1+e^x}$. T011V covers slides 4–10; T011Vb covers 11–17 and then works a Cauchy equation that is on no slide (W2.4, transcript only). CW P23–P28 and the four ESA questions are solved in Topics 3–4.